

ADVANCED DATA ANALYSIS & BUSINESS INTELLIGENCE

Business Insights & Recommendations Report

Superstore Sales & Profit Performance Analysis

Prepared for: Week 3 Advanced Data Analysis Assignment

Dataset: Superstore Sales Dataset

Tool Used: Microsoft Power BI

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Executive Summary

This report presents the findings of an advanced analysis of Superstore sales and profitability performance using Microsoft Power BI. The analysis builds on the Week 2 project and extends the investigation through KPI development, DAX measures, time-based analysis, discount and profitability analysis, product-level performance and customer analysis.

The results show positive sales growth over the period analysed, with 2017 recording the strongest sales performance at approximately \$0.73 million. However, the analysis also identifies important profitability risks. Certain products and sub-categories generate weak or negative profit, while the discount analysis indicates that moderate to high discounts may contribute to reduced profitability.

The analysis highlights uneven performance across regions and categories. The West region was the strongest performer, while the South generated the lowest sales. Technology was the strongest category for sales and profit, whereas Furniture produced comparatively weak profit. At product level, the Canon ImageClass 2200 Advanced Copier generated the highest sales at R61,599.82, while the Zebra GK420t Direct Thermal/Thermal Transfer Printer recorded a loss of R938.28. Sean Miller was identified as the highest-value customer in the analysis, with sales of R25,043.05.

The key recommendation is for management to complement sales growth targets with stronger profitability monitoring. Discount policies, loss-making products, weaker categories and regional performance should be reviewed regularly to ensure that revenue growth is translated into sustainable profit.

1. Analysis Objective and Approach

The objective was to convert transactional sales data into meaningful business insights that can support management decision-making. The analysis focused on:

- Sales and profit trends over time
- Profitability by category and sub-category
- The relationship between discounting and profitability
- Product-level sales and profit performance
- Regional and customer performance
- Business problems, opportunities and practical recommendations

2. KPI and DAX Development

Eight DAX measures were developed to support the analysis and dashboard interactivity.

Measure	Business Purpose
Total Sales	Measures total revenue generated.
Total Profit	Measures overall profitability.
Total Orders	Counts unique orders.

Total Customers	Counts unique customers.
Profit Margin	Measures profit relative to sales.
Average Sales per Order	Measures average revenue per order.
Average Discount	Measures the average discount applied.
Loss-Making Orders	Identifies orders associated with negative profit.

3. Key Business Problems Investigated

3.1 Discounting and Profitability Risk

The discount versus profit analysis was undertaken to assess whether discounting may be contributing to weak or negative profitability. The visual analysis showed several observations below the zero-profit line at moderate to high discount levels. The Profit by Sub-Category analysis also showed that profitability is not consistent across the product portfolio.

This suggests that increasing sales through discounting does not automatically result in improved business performance. Where discounts are applied without sufficient consideration of product margin, sales growth can be achieved at the expense of profitability.

3.2 Product-Level Profitability Risk

The product performance analysis identified a significant difference between high sales performance and actual profitability. The Canon ImageClass 2200 Advanced Copier was the highest-selling product at R61,599.82. In contrast, the Zebra GK420t Direct Thermal/Thermal Transfer Printer recorded a negative profit of R938.28.

This demonstrates that sales performance alone is not a sufficient measure of business success. Individual products can contribute substantial revenue while other products reduce overall profitability. Product-level profitability should therefore be monitored alongside sales performance.

3.3 Uneven Regional and Category Performance

The analysis identified material differences in performance across regions and categories. The West region was the strongest overall performer, while the South recorded the lowest sales. Technology was the strongest category for sales and profit, whereas Furniture produced comparatively weak profit.

These differences indicate that performance is not evenly distributed across the business. Management should investigate the drivers of strong performance and determine whether successful practices can be replicated in weaker regions or categories.

4. Key Business Insights

Sales growth strengthened over time: Sales increased from approximately \$0.48 million in 2014 to approximately \$0.73 million in 2017. After a slight decline in 2015, sales recovered and grew strongly in 2016 and 2017.

Profit generally followed the sales growth trend: Total profit also improved over the period, indicating that overall growth was not limited to revenue. Product and category analysis, however, shows that profitability remains uneven and should not be assessed only at total-company level.

Discounting presents a potential profitability risk: The discount and profit analysis showed several observations with negative profit at moderate to high discount levels. This indicates a need for more controlled discount decisions, particularly for products with limited margins.

Technology is a major contributor to performance: Technology generated the strongest category performance in both sales and profit. This category represents an important source of value and should remain a strategic focus while maintaining appropriate margin controls.

Product performance varies significantly: The highest-selling product generated R61,599.82 in sales, while another product generated a loss of R938.28. This highlights the importance of monitoring both revenue and profitability at product level.

High-value customers represent an opportunity: Sean Miller was identified as the top customer in the analysis, generating R25,043.05 in sales. Identifying and retaining high-value customers can support revenue stability and growth.

Regional performance is uneven: The West was the strongest region, while the South generated the lowest sales. This creates an opportunity to investigate regional differences and address factors contributing to weaker performance.

5. Recommendations

Introduce profitability-based discount controls

Discounts should be reviewed against product margin before approval. Management should consider discount thresholds or additional approval where proposed discounts may result in low or negative profitability.

Establish a product profitability review process

Regular reports should identify loss-making products and products with high sales but weak margins. The underlying causes should be investigated, including pricing, discounting, costs and product mix.

Review underperforming categories and sub-categories

Categories or sub-categories with consistently weak profit should be reviewed to determine whether pricing, supplier costs, promotions or product strategy require adjustment.

Replicate successful regional practices

Management should analyse the factors contributing to the West region's stronger performance and determine whether successful sales, operational or product strategies can be adapted for weaker regions, particularly the South.

Strengthen high-value customer management

High-value customers should be monitored and supported through appropriate relationship management and service strategies. Customer analysis can also identify other customers with the potential to increase their contribution.

Use the dashboard as an ongoing decision-support tool

The Power BI dashboard should be refreshed and reviewed regularly so that management can monitor changes in sales, profit, discounting and customer performance rather than relying only on periodic historical analysis.

6. Conclusion

The analysis demonstrates that the Superstore business achieved positive sales and profit growth over the period reviewed. However, the results also show that overall growth can conceal important profitability risks at product, sub-category and regional level. The strongest opportunity for improvement lies in strengthening profitability management by combining revenue analysis with product-level margin monitoring and more disciplined discount controls.

The Power BI dashboard provides a practical platform for management to monitor these areas interactively. By acting on the identified recommendations, management can focus not only on increasing sales, but also on improving the quality and sustainability of profit.

Appendix A: Evidence Captured from the Dashboard

Metric	Finding
Highest sales year	2017-approximately \$0.73M
Lowest sales year	2015-approximately \$0.46M
Top product by sales	Canon ImageClass 2200 Advanced Copier-R61,599.82
Lowest identified product profit	Zebra GK420t Direct Thermal/Thermal Transfer Printer-R938.28
Top customer by sales	Sean Miller-R25,043.05
Strongest region	West
Lowest sales region	South
Strongest category	Technology
Weak category profitability	Furniture
Largest customer segment	Consumer

